071306T4EEN

Electrical Engineering (Power Option) Level 6
ENG/OS/PO/CR/10/6

Demonstrate Understanding of Power Generation

July/August 2023

Time: 3hours

TVET CURRICULUM DEVELOPMENT, ASSESSMENT AND CERTIFICATION COUNCIL (TVET CDACC)

WRITTEN ASSESSMENT GUIDE

INSTRUCTIONS TO ASSESSOR

*In this assessment, the candidate will be required to answer **written** questions;*

Marks for each question are indicated in the brackets

*The paper consists of **TWO** sections: **A and B***

This guide consists of ten (10) printed pages.

Assessor should check the guide to ascertain that all the pages are printed as indicated and that no questions or answers are missing.

SECTION A (40 MARKS)

1. Define the term power generation. (2 Marks)

   It is the conversion of energy available in different forms in nature into electrical energy. *(Award 2 or 0 marks for correct definition)*

2. Name **five** sources of energy. (5 Marks)

   i. **Sun**
   ii. **Wind**
   iii. **Water**
   iv. **Fuels**
   v. **Nuclear energy.**
   vi. **Tide**

   (Award 1 or 0 mark for any correct answer)

3. Name any **four** types of generating station. (4 Marks)

   i. **Steam power station (thermal station)**
   ii. **Hydro-electric power station**
   iii. **Diesel power station**
   iv. **Nuclear power station**
   v. **Magnetohydrodynamic (MHD) generation**
   vi. **Gas Turbines power plant** *(Award 1 mark for any correct answer)*

4. List any **six-**power generation station in Kenya. (6 Marks)

   i. **Kamburu hydro power station**
   ii. **Kindaruma hydro power station**
   iii. **Gitaru hydro power station**
   iv. **Kiambere hydro power station**
   v. **Turkwel hydro power station**
   vi. **Olkaria geothermal power station**
   vii. **Masinga hydro power station (***Award 1 or 0 mark for any correct answer)*

5. Define the term **PPE** as used at work and give two examples of PPEs commonly used while working on a power generation station. (4 Marks)

   It is all equipment designed to be worn, or held, to protect against a risk to health and safety.

They include safety helmets, Googles for eye protection, Dust coat/Overall, safety boots. *(Award 4 or 0 marks for correct definition)*

6. List **four** reasons for using the grid system from power generation station. (4 Marks)
    i. **Stabilizing the power supply i.e., when one generator fails, the other can supply power**
    ii. **Ensure stability in the frequency of the power supply**
    iii. **Use of older plants and less efficient plants to carry peak loads for a short duration**
    iv. **Exchange of peak loads**
    v. **Reduce plant reserve capacity**
    vi. **Ensure economical operation – arrangement so that more efficient stations work continuously throughout the year at a high load factor.** *(Award 1 or 0 mark for any correct answer)*
7. List **four** advantages of diesel power station compared with hydro-power station. (4 Marks)
    i. **It occupies less space**
    ii. **It can be located anywhere**
    iii. **It requires less quantity of water**
    iv. **The design and layout of the plant are quite simple**
    v. **It requires less operating staff** *(Award 1 mark for each correct answer)*
8. Name **three** electrical equipment of a hydro-electric power station. (3 Marks)
    i. **Alternator**
    ii. **Transformers**
    iii. **Circuit breakers**
    iv. **Switching devices** *(Award 1 or 0 mark for any correct answer)*
9. Name **four** main parts of Magneto hydrodynamic (MHD) generation. (4 Marks)
    i. **Cathode wall**
    ii. **Anode wall**
    iii. **Insulating wall**
    iv. **Magnet (Magnetic field)**
    v. **Working fluid (Plasma)** *(Award 1 mark for each correct answer)*
10. Define the term thermal efficiency of a steam power station. (2 Marks)

The ratio of heat equivalent of mechanical energy transmitted to the turbine shaft to the heat of combustion of coal is known as thermal efficiency of steam power station. *(Award 2 or 0 for correct definition)*

i.e $\eta_{\text{thermal}} = \dfrac{\textbf{Heat equivalent of mech.energy transmitted to turbine shaft}}{\textbf{Heat of coal combustion}}$

11. Name **two** types water turbines mainly used in hydro-power plant. (2 Marks)

    i. **Pelton wheel turbine**
    ii. **Francis turbine**
    iii. **Kaplan turbine** *(Award 1 or 0 mark for any correct answer)*

SECTION B (60 MARKS)

Attempt any THREE questions (All questions carry equal marks)

12. a) Describe the use of the following parts in a steam generating station. (12 Marks)

    i. **Coal and ash handling plant.** The coal is transported and stored in the coal storage plant then is delivered to the coal handling plant where it is pulverized (i.e., crushed into small pieces) in order to increase its surface exposure.
    ii. **Boiler**. It is a closed vessel in which water is converted into steam by utilizing the heat combustion of coal at high temperature and pressure.
    iii. **Economizer.** It is a device which heats the feed water on its way to the boiler by deriving heat from flue gases. This result in raising boiler efficiency.
    iv. **Air preheater**. It extracts heat from flue gases and increases the temperature of air used for coal combustion to increase thermal efficiency and steam capacity.
    v. **Steam turbine**. It is used for converting steam energy when passing over the blades of turbine into mechanical energy.
    vi. **Alternator**. The steam turbine is coupled to an alternator. The alternator converts mechanical energy of turbine into electrical energy. The electrical output from the alternator is delivered to the bus bars through transformer, circuit breakers and isolators.

    (Award 2 marks for each correct answer)

    b) KENGEN is planning to start a new hydro-electric power generation. You are provided with the following data;

    Catchment area = 5×10^9 m^2 ;

Mean head, H = 30 m

Annual rainfall, F = 1·25 m;

Yield factor, K = 80 %

Overall efficiency, η overall = 70 %

Calculate the average power in kW that can be generated and determine the appropriate rating in kW of generators to be installed putting in consideration load factor is 40%. (8 marks)

Solution

Average power available = Weight of water (W) available x Mean head (H) x Overall efficiency (η) over the hours available in a year

i. **Weight of water (W) available per year will be;**

Volume of water x Gravitational force

ii. **Volume of water which can be utilized per year will be;**

Catchment area x Annual rainfall x Yield factor

(5×10^9) m^2 x (1·25) m x (0.8) = 5 x 10^9 m^3

Thus; W = (5×10^9) m^3 x (9.81) x (1000) = 49.05 x 10^{12} N (Award 2 marks)

iii. **Electrical Energy available per year will be;**

Weight of water (W) available x Mean head (H) x Overall efficiency (η)

$$\mathbf{W \times H \times \eta} = \frac{(49.05 \times 10^{12}) \times (30) \times (0.7)}{1000 \times 3600} \text{ kWh}$$

= 2.86 x 10^8 kWh (Award 2 marks)

Therefore, Average power = Electrical Energy available over the hours available in a year (8760)

$$= \frac{2.86 \times 10^8}{8760} \text{ kW}$$

= 32648 kW (Award 2 marks)

iv. **Maximum capacity of the generator required = Maximum demand**

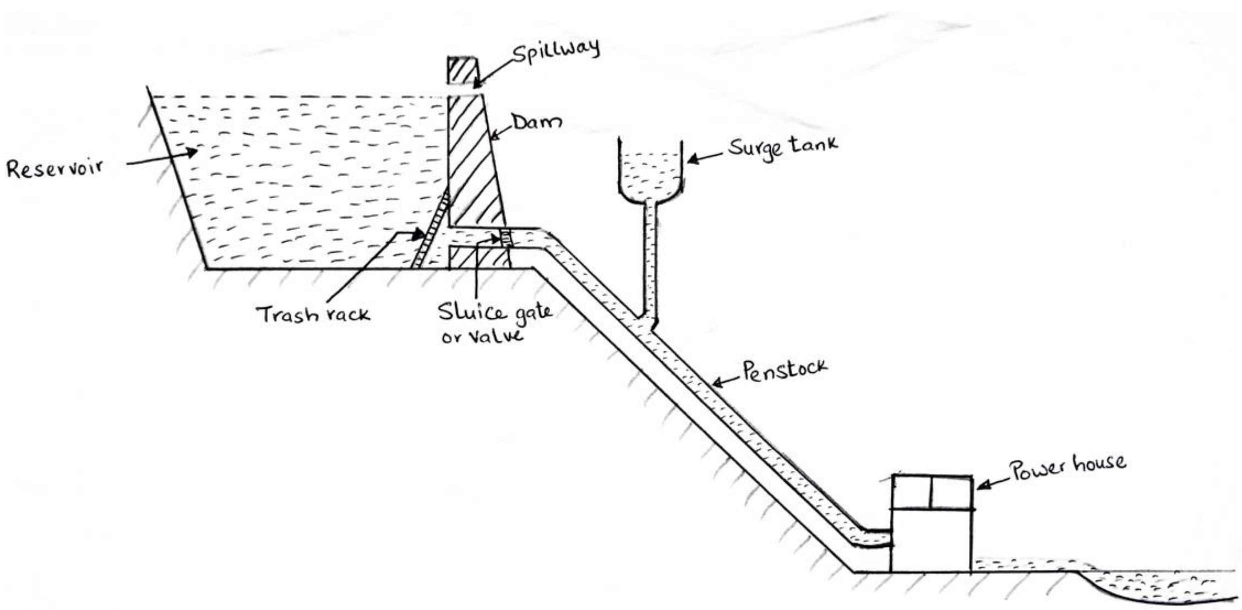
v. **Maximum demand is given by the ratio of average power to the load factor.**

$$\text{Max. demand} = \frac{\text{Average demand}}{\text{Load factor}}$$

$$= \frac{32{,}648}{0.4} = 81620 \text{ kW}$$ **(Award 2 marks)**

13. a) Using a labelled schematic arrangement of hydro-electric power station, explain the function of the following parts. (12 Marks)

i. Dam
ii. Spillways
iii. Surge tank
iv. Penstock



i. **Dam**. A barrier which stores water and creates water head. The type and arrangement depend upon the topography of the site.

ii. **Spillways**. They are used to discharge the surplus water from the storage reservoir into the river on the down-stream side of the dam. Gates are provided between these piers and surplus water is discharged over the crest of the dam by opening these gates.

iii. **Surge tank**. It is a small reservoir or tank (open at the top) in which water level rises or falls to reduce the pressure swings in the conduit. A surge tank is located near the beginning of the conduit.

iv. **Penstocks**. They are open or closed conduits which carry water to the turbines. various devices such as automatic butterfly valve, air valve and surge tank are provided for the protection of penstocks. Automatic butterfly valve shuts off water flow through the penstock promptly if it ruptures.

(Award 4 marks for correct diagram, Award 2 marks for each correct explanation)

b) A steam power station spends, Ksh 3,000,000 per year for coal used in the station. The coal has a calorific value of 5000 Kcal/kg and cost Ksh 300 per ton. If the station has a thermal efficiency of 33% and electrical of 90%, find the average load on the station. (8 Marks)

Solution:

Average load on the station is given by the ratio of units generated per year to the hours available in a year.

i. **Units generated per year =** $\frac{\textbf{Heat output}}{\textbf{Equivalent electrical heat (860Kcal)}}$**;NB 1Kwh = 860Kcal**

ii. **Heat output = Overall efficiency x Heat of combustion**

iii. **Heat of combustion = Coal used x Calorific value**

iv. **Overall efficiency = Thermal efficiency x Electrical efficiency**

= 0.33 x 0.9 = 0.27

Coal used per year $= \frac{3 \times 10^6}{300} = 10 \times 10^6$ **Kg**

Heat of combustion $= 10 \times 10^6$ **Kg x 5000** $= 5 \times 10^{10}$ **Kcal (Award 2 marks)**

Heat output $= 0.27 \times 5 \times 10^{10}$ **Kcal** $= 1487 \times 10^7$ **Kcal (Award 2 marks)**

Units generated per year $= \frac{1487 \times 10^7}{860}$ $= 17.3 \times 10^6$**Kwh (Award 2 marks)**

Average load on station $= \frac{17.3 \times 10^6}{8760}$ **= 1974 Kw (Award 2 marks)**

14. a) Describe the following plant auxiliaries of a diesel power station. (12 Marks)

i. Fuel supply system. **It consists of storage tank, strainers, fuel transfer pump and all-day fuel tank. This oil is stored in the storage tank. From the storage tank, oil is pumped to smaller all-day tank at daily or short intervals. From this tank, fuel oil is passed through strainers to remove suspended impurities. The clean oil is injected into the engine by fuel injection pump.**

ii. Air intake system. **This system supplies necessary air to the engine for fuel combustion. It consists of pipes for the supply of fresh air to the engine manifold. Filters are provided to remove dust particles from air which may act as abrasive in the engine cylinder.**

iii. Cooling system. **This system keeps the temperature of the engine parts within the safe operating limits. The cooling system consists of a water source, pump and cooling towers. The pump circulates water through cylinder and head jacket. The water takes away heat form the engine. The hot water is cooled by cooling towers and is recirculated for cooling.**

iv. Lubricating system. **This system minimizes the wear of rubbing surfaces of the engine. It comprises of lubricating oil tank, pump, filter and oil cooler. The lubricating oil is drawn from the lubricating oil tank by the pump and is passed through filters to remove impurities. The clean lubricating oil is delivered to the points which require lubrication. The oil coolers incorporated in the system keep the temperature of the oil low.** ***(Award 3 or 0 for correct answer)***

b) Kenya expects to start generating electricity from nuclear power by the year 2030. As an engineer delegated in planning, explain **four** factors considered when selecting the site for a nuclear power station. (8 Marks)

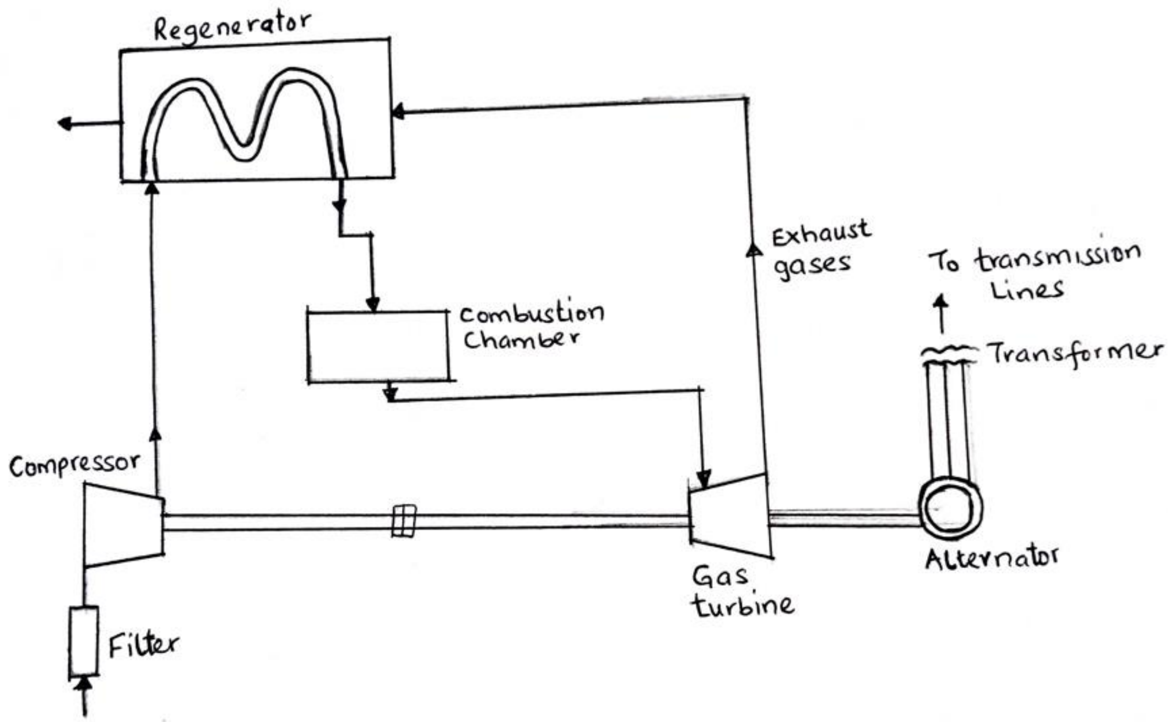
i. Availability of water. The site should be located where ample plenty of water is available such as near a river for cooling purposes.

ii. Disposal of waste. Nuclear station produces waste which are generally radioactive and dangerous. Such waste should be disposed by burring in deep trench where cannot be hazardous to environment, or underground water.

iii. Distance from populated areas. The site should be away from populated areas as there's a danger of presence radioactivity in the area near the generating station.

iv. **Transportation facilities. The site must be located in area where it can be easily assessed for transportation of equipment required in the site as well as facilitate the movement of workers.** *(Award 2 or 0 for any correct answer)*

15. a) Having trained as engineer, you have been appointed to explain to trainees how gas generation station works, using a well labelled schematic diagram explain the main parts of the gas generation station. (12 Marks)



i. **Compressor**. **The compressor used in the plant is generally of rotatory type. The air at atmospheric pressure is drawn by the compressor via the filter which removes the dust from air. The rotatory blades of the compressor push the air between stationary blades to raise its pressure. Thus, air at high pressure is available at the output of the compressor.**

ii. **Regenerator. A regenerator is a device which recovers heat from the exhaust gases of the turbine. It consists of a nest of tubes contained in a shell. The exhaust is passed through the regenerator before passed to atmosphere. The compressed air from the compressor passes through the tubes on its way to the combustion chamber where it is heated by the hot exhaust gases thus raising its pressure.**

iii. **Combustion chamber. The air at high pressure from the compressor is led to the combustion chamber via the regenerator. In the combustion chamber, heat is added to the air by burning oil. The oil is injected through the burner into the chamber at high pressure to ensure atomization of oil and its thorough mixing with air. The result is that the chamber attains a very high temperature (about 3000º F). The combustion gases are suitably cooled to 1300o F to 1500o F and then delivered to the gas turbine.**

iv. **Gas turbine. The products of combustion consisting of a mixture of gases at high temperature and pressure are passed to the gas turbine. These gases in passing over the**

turbine blades expand and thus do the mechanical work. The temperature of the exhaust gases from the turbine is about 900^{o} F

v. **Alternator. The gas turbine is coupled to the alternator. The alternator converts mechanical energy of the turbine into electrical energy. The output from the alternator is given to the bus-bars through transformer, circuit breakers and isolators.**

(Award 4 marks for correct diagram, Award 2 marks for each correct explanation)

b) Being an engineer in energy sector you are invited for power generation seminar. You have been delegated to talk about renewable power generation. Highlight **five** advantages and **three** disadvantages of solar energy power generation. (8 Marks)

Advantages.

i. **Solar power is pollution free**
ii. **Reduces dependence on oil and fuels**
iii. **It available everyday**
iv. **No monthly bills**
v. **It can be installed anywhere**
vi. **Solar energy can be stored** *(Award 1 or 0 mark for any correct answer)*

Disadvantages

i. **High initial cost for materials and installation**
ii. **Its weather dependent**
iii. **Solar energy storage is expensive**
iv. **Uses a lot of space for large production of power** *(Award 1 or 0 mark for any correct answer)*

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